Cheatsheet AZ-305 — Azure SQL (PaaS): HA and DR

Applies to **Azure SQL Database** and **Azure SQL Managed Instance**. Azure manages the engine; you choose the mechanism.

1. The four mechanisms in one table

	Zone-redundant HA	Active geo-replication	Auto-failover group	Geo-restore
Protects against	Node / zone (data center) failure	Region outage	Region outage	Region outage (data only)
Reach	Within the region	Across regions (or same)	Across regions	Across regions
Replication	Synchronous	Asynchronous	Asynchronous	Backup copy
RPO	**0**	Seconds	Seconds	Up to 1 hour
RTO	Seconds	Seconds-minutes	Seconds-minutes	**Hours**
Failover	Automatic	**Manual**	**Automatic** (or manual)	Manual (restore)
Connection string	Does not change	**Changes** (different server)	**Does not change** (listener)	Changes (new server)
Scope	1 database	1 database	**Several DBs / whole MI**	1 database
Readable secondaries	Yes on Business Critical / Premium	Up to **4**, readable	**1**, readable (.secondary)	No
Managed Instance	Yes	**No**	**Yes** (the only geo-DR for MI)	Yes
Cost	Included in the tier	Each secondary is paid	The secondary is paid	Backup storage only

2. Zone-redundant HA (local high availability)

- It is not DR: replicas within the **same region**, spread across availability zones.
- **General Purpose zone-redundant:** ZRS storage + relocatable compute. Survives 1 zone.
- **Business Critical / Premium (DTU):** internal Always On AG, **primary + 3 synchronous secondaries**, local SSD. Survives **2 zones** down. Read replica included.
- **Standard (DTU) and Basic: do not** support zone redundancy.
- Signals: "zone outage", "data center failure", "no data loss", "minimize I/O latency".

3. Active geo-replication

- Creates up to 4 **readable secondaries** in other regions, continuous asynchronous replication.
- **You** decide the failover (or your script). When promoting, the app must point to the **new logical server**.
- Only **Azure SQL Database**, one DB per configuration.
- Signals: "readable secondaries", "manual failover", "a single database", "distribute reads across regions".
- Trap: if the wording says "without administrative intervention" → it is **not** this one.

4. Auto-failover group

- Active geo-replication + automation + connection abstraction.
- Groups **several DBs of a logical server** (or a whole MI): they fail over **together**.
- Two fixed **listener endpoints**: .database.windows.net → read/write (current primary) .secondary.database.windows.net → read-only
- **Automatic** failover with configurable grace period, or manual.
- **1 secondary**, mandatorily in **another region**.
- It is the **only** geo-DR mechanism for **Managed Instance**.
- Signals: "automatic failover", "without administrative intervention", "without modifying the app / connection string", "several DBs must fail over together", "Managed Instance".

5. Geo-restore

- Automatic backups (weekly full, diff every 12-24 h, log every 5-10 min) copied to the paired region (RA-GRS).
- On disaster: **create a new DB** by restoring the latest available backup.
- Cheap, but **RTO of hours** and **RPO of up to 1 h**.
- Signals: "minimize costs", "recover the data", tolerates hours of downtime.
- Trap: it is never the answer to "minimize downtime".

6. Firewall and failover groups (exam detail)

- **Database-level** rules (sp_set_database_firewall_rule) → **travel with the replica** on failover.
- **Server-level** rules → are **not** replicated to the secondary server.
- Symptom "one DB connects after the failover and another gets a firewall error" → mix of levels.

7. How they combine (typical mission-critical design)

```
Business Critical zone-redundant  (local HA, RPO 0, automatic)
        +
Auto-failover group to another region  (DR, RPO seconds, automatic)
        +
Automatic backups               (last safety net: corruption / deletion, which replication does NOT protect against)
```

8. Decision tree

```
What must survive?
├─ Node / zone ..................... Zone-redundant (Premium / Business Critical)
├─ Region, manual control, reads ... Active geo-replication
├─ Region, automatic, app untouched  Auto-failover group
├─ Region, data only, minimum cost . Geo-restore
└─ It is Managed Instance .......... Auto-failover group (there is no other)
```

9. Signals → answer

Wording	Answer
"zone outage" + "no data loss"	Zone-redundant / Premium / Business Critical
"readable secondaries in multiple regions"	Active geo-replication
"manual failover"	Active geo-replication
"automatic failover" / "without admin intervention"	Auto-failover group
"connection string must not change" / "minimize changes to the app"	Auto-failover group
"multiple databases fail over together"	Auto-failover group
"Managed Instance" + regional DR	Auto-failover group
"minimize costs" + tolerates hours	Geo-restore
"restore to another region"	Geo-restore

10. Mistakes to avoid

1. Believing that active geo-replication does automatic failover → **no**, that is the failover group.
2. Choosing geo-restore when asked to minimize downtime → RTO of hours.
3. Choosing Standard (DTU) for zone redundancy → **it does not support it**.
4. Choosing active geo-replication for Managed Instance → **it does not exist** on MI.